Proposal: Characterizing Deception in Large Language Models via Latent Spectral Dynamics

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Abstract

As the capabilities of Large Language Models (LLMs) scale, alignment failures such as strategic deception and sycophancy present critical safety risks. Current mechanistic interpretability approaches largely analyze these models statically, missing the temporal evolution of hidden states during autoregressive generation. In this work, we propose extending the *Latent Spectral Dynamics* framework to characterize deception. Building upon its recent application in analyzing hallucinations, we model the token-wise residual stream as a stochastic, autonomous, discrete-time dynamical system. We lift these nonlinear dynamics into an infinite-dimensional linear space using Koopman operator theory. To overcome the computational bottleneck of our current matrix-factorization-based Hankel-EDMD approach, we propose a deep learning paradigm, training Deep Koopman Autoencoders to scale spectral analysis to larger models. Ultimately, we hypothesize that the Markovian transition in the dynamics will persist and strategically deceptive trajectories will exhibit distinct spectral signatures, distinguishing them from both factual and uncertain generation.

1 Introduction and Motivation

Large Language Models (LLMs) are increasingly deployed as autonomous agents and reasoning engines. However, this increased capability brings sophisticated alignment failures. While *hallucination* (generating fluent but factually ungrounded text) is well-documented, a more insidious failure mode is *deception*. Deceptive models may strategically output false information to achieve a goal (Scheurer et al., 2023), hide their true capabilities to evade safety protocols (Hubinger et al., 2024), or sycophantically align with user biases despite retaining internal knowledge of the truth.

The primary motivation for this project is to characterize these emergent deceptive behaviors through the lens of computational dynamical systems. This project extends upon our current work in Mechanistic Interpretability (currently under review for COLM 2026), where we successfully formulated LLM autoregressive generation as an autonomous discrete-time dynamical system. In that work, we utilized Koopman operator theory and Hankel-Extended Dynamic Mode Decomposition (Hankel-EDMD) to demonstrate that factual and hallucinated regimes possess distinctly identifiable spectral signatures on the approximated Koopman operators.

However, Hankel-EDMD relies on exact matrix factorizations over massive, uncompressed coordinate bases, making it prohibitively expensive in CPU and RAM when scaling to larger models (e.g., 32B+ parameters). Furthermore, deception is a fundamentally more complex phenomenon than hallucination; it requires the model to maintain a dual-state representation (the hidden truth vs. the deceptive output).

Our objectives for this course project are thus threefold:

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1. *Algorithmic Extension:* We aim to bypass the EDMD matrix factorization bottleneck by implementing a Deep Koopman Autoencoder to learn the spectral dynamics efficiently.
2. *Phenomenological Extension:* We will apply this upgraded framework to datasets designed to invoke strategic deception, analyzing the global latent dynamics to identify the unique spectral motif of deceptive alignment.
3. *Narrative Shift:* We will develop the theoretical language to interpret deception formally from a dynamics angle—framing it as a geometrically distinct attractor or invariant subspace within the activation manifold.

The proposer promises heavy individual effort throughout this extension project.

2 Related Work

This project bridges in-context learning, mechanistic interpretability, and dynamical systems:

1. **Implicit Bayesian Inference in ICL:** Xie et al. (2022) formalized In-Context Learning (ICL) as implicit Bayesian inference, suggesting that sufficiently prompted LLMs operate over latent concepts, justifying our locally Markovian assumptions.
2. **Strategic Deception in LLMs:** Scheurer et al. (2023) demonstrated that LLMs can strategically deceive users (e.g., insider trading scenarios) when put under pressure, providing the behavioral framing for our work.
3. **Sleeper Agents:** Hubinger et al. (2024) highlighted the persistent danger of deceptive alignment, showing that models can harbor hidden deceptive behaviors that evade standard safety fine-tuning.
4. **Universal Motifs in LLM Deception:** Yang and Buzsaki (2025) recently characterized deception across multiple LLM families, identifying a universal motif consisting of three iterative refinement stages in the latent representation. Their success in controlling outputs by steering these specific stages strongly supports our hypothesis that deception manifests as a distinctly identifiable temporal and geometric structure within the activation manifold.
5. **Deep Koopman Autoencoders:** Lusch et al. (2018) introduced a deep learning framework for discovering universal linear embeddings of nonlinear dynamics, which we adapt to replace our algebraic Hankel-EDMD solver.

3 Methodology

This framework shifts the paradigm of LLM autoregressive generation by framing it as a stochastic, autonomous dynamical system, providing a mathematically grounded approach to diagnosing behavioral divergence (such as deception) in generative trajectories using spectral signatures.

3.1 In-Context Learning and Phase Space Reconstruction

Autoregressive generation is typically viewed as a highly non-Markovian joint probability $P_\phi(x_{1:N}) = \prod_{k=1}^{N} P_\phi(x_k \mid x_{<k})$. Motivated by Xie et al. (2022), we assume generation relies on a latent concept $\theta \in \Theta$, rendering the process *locally Markovian*: $P_\phi(x_{k+1} \mid x_{1:k}, \theta) = P_\phi(x_{k+1} \mid x_k, \theta)$. Marginalizing over the LLM's implicit inference of θ yields:

$$P_\phi(x_{k+1} \mid x_{1:k}) = \int_\Theta P_\phi(x_{k+1} \mid x_k, \theta) P_\phi(\theta \mid x_{1:k}) d\theta \tag{1}$$

To construct an approximately Markovian phase space, we apply Hankel Time-Delay Embedding to the final-layer residual stream $h_k \in \mathbb{R}^d$, forming $\mathbf{H}_k = [h_{k-W+1}^\top, \ldots, h_k^\top]^\top \in \mathbb{R}^{Wd}$. $\mathbf{H}_k$ acts as a *sufficient statistic* for the sequence history, fully parameterizing the implicit posterior $P_\phi(\theta \mid x_{1:k}) \equiv P_\phi(\theta \mid \mathbf{H}_k)$. If a directive prompt collapses this posterior onto a target concept θ^*, generation becomes strictly dependent on the embedded state: $P_\phi(x_{k+1} \mid x_{1:k}) \approx P_\phi(x_{k+1} \mid \mathbf{H}_k, \theta^*)$.

3.2 Autonomous Dynamics Formulation

To resolve the type mismatch between discrete tokens $x_k \in \mathcal{V}$ and continuous states $\mathbf{H}_k \in \mathbb{R}^{Wd}$, we apply the *Gumbel-Max reparameterization trick*. This maps the temperature-scaled categorical sampling into a deterministic function driven by exogenous noise $\mathbf{\Xi}_k \sim \text{Gumbel}(0, \mathbf{I}_{|\mathcal{V}|})$: $x_{k+1} \equiv \mathcal{S}_{\tau,\theta^*}(\mathbf{H}_k, \mathbf{\Xi}_k)$. Substituting this emission map directly into the Transformer's deterministic update $h_{k+1} = f_\phi(\mathbf{H}_k, x_{k+1})$ eliminates the discrete token entirely, yielding a **stochastic autonomous discrete-time dynamical system**:

$$\mathbf{H}_{k+1} = \mathcal{F}_{\phi,\theta^*}(\mathbf{H}_k, \mathbf{\Xi}_k) \tag{2}$$

The sequence $\{\mathbf{H}_k\}$ forms a Markov chain on $\mathbb{R}^{Wd}$, governed by the **Markov Transition Kernel**:

$$\mathcal{P}_{\theta^*}(d\mathbf{H}_{k+1} \mid \mathbf{H}_k) = \mathbb{P}_{\mathbf{\Xi}}\left(\mathcal{F}_{\phi,\theta^*}(\mathbf{H}_k, \mathbf{\Xi}_k) \in d\mathbf{H}_{k+1}\right) \tag{3}$$

(Note: In the limit of greedy decoding as $\tau \to 0^+$, the stochastic forcing collapses, reducing the system to a purely deterministic autonomous system).

3.3 The Koopman Operator and Deep Learning Extension

We lift the dynamics into an infinite-dimensional Hilbert space via the Koopman operator $\mathcal{K}_{\theta^*}$, defined by conditional expectation over the Markov kernel: $(\mathcal{K}_{\theta^*} g)(\mathbf{H}_k) = \mathbb{E}_{\mathbf{H}_{k+1} \sim \mathcal{P}_{\theta^*}(\cdot|\mathbf{H}_k)}[g(\mathbf{H}_{k+1})]$.

While our prior work approximated $\mathcal{K}_{\theta^*}$ via Hankel-EDMD, the Gram matrix construction limits scaling. Thus, we propose replacing EDMD with a **Deep Koopman Autoencoder** (Lusch et al., 2018). We define an encoder $\psi : \mathbb{R}^{Wd} \to \mathbb{R}^m$, decoder $\phi : \mathbb{R}^m \to \mathbb{R}^{Wd}$, and a learned Koopman matrix $\mathbf{K} \in \mathbb{R}^{m \times m}$, optimized via:

$$\mathcal{L} = \underbrace{\|\mathbf{H}_k - \phi(\psi(\mathbf{H}_k))\|^2}_{\text{Reconstruction}} + \alpha \underbrace{\|\psi(\mathbf{H}_{k+1}) - \mathbf{K}\psi(\mathbf{H}_k)\|^2}_{\text{Linear Dynamics}} + \beta \|\mathbf{H}_{k+1} - \phi(\mathbf{K}\psi(\mathbf{H}_k))\|^2 \tag{4}$$

This parametric approach decouples sequence length from memory constraints, enabling scalable extraction of deceptive spectral properties (eigenvalues and invariant subspaces).

4 Feasibility and Experimental Design

Evidence of Understanding & Extension Plan: This proposal builds directly on an existing codebase and theoretical framework co-authored by the proposer. Feasibility is high as the base HuggingFace extraction pipeline and delay-embedding infrastructure are already validated. We will execute a modular shift: swapping the algebraic EDMD solver for a Deep Koopman Autoencoder, and shifting our focus from standard hallucination datasets to strategic deception.

Data, Models, & Generalization: We will adapt the Apollo Research "Insider Trading" dataset (Scheurer et al., 2023) to induce strategically deceptive continuations. To rigorously test the universality of the deceptive motif, we will additionally evaluate generalization on a Sycophancy dataset to verify if the spectral signature is consistent across different modes of deception. We will extract final-layer residual stream trajectories ($\mathbf{H}_k$, $W = 5$) from open-weight models (Qwen2.5-14B/32B/72B) across a temperature gradient.

Training & Ablations: The Autoencoder will be optimized using AdamW. We will perform strict ablations on the linear transition loss penalty (α) to guarantee that the network learns true temporal dynamics rather than merely memorizing static state representations.

Evaluation Metrics & Baselines: We will evaluate the framework via:

1. *Predictability:* We will measure multi-step rollout Mean Squared Error (MSE) on held-out trajectories (i.e., $\mathbf{H}_{k+n} \approx \phi(\mathbf{K}^n \psi(\mathbf{H}_k))$) to ensure the Koopman matrix $\mathbf{K}$ accurately captures the autonomous n-step Markovian dynamics.
2. *Spectral Signatures:* Eigendecomposition of $\mathbf{K}$ to extract eigenvalues λ and spatial modes. Orthogonal eigenmodes will verify that deception occupies a distinctly separable invariant subspace within the activation manifold.

3. *Static Probing Baselines:* We will benchmark our spectral anomaly detection against standard static probing (e.g., training a linear logistic regression classifier on isolated snapshots h_k). We hypothesize that tracking the dynamic Koopman eigenvalues will identify deceptive trajectories earlier and more robustly out-of-distribution compared to static classification.

5 Project Planning

- **Week 2:** *Setup & Baselines.* Adapt the insider trading dataset. Extract final-layer trajectories for truthful vs. deceptive responses using the existing pipeline (Qwen2.5-14B).
- **Week 3–4:** *Deep Learning Paradigm.* Implement and train the Deep Koopman Autoencoder. Validate that it achieves multi-step rollout MSE comparable to our previous EDMD baseline.
- **Week 5:** *Scale & Analyze Deception.* Run the pipeline across larger models (Qwen2.5-32B/72B). Extract the eigenvalues λ and spatial modes from the learned matrix $\mathbf{K}$.
- **Week 6:** *Spectral Comparison.* Quantify the difference between truthful and deceptive regimes (e.g., examining if deceptive trajectories manifest as high-magnitude complex-conjugate eigenvalues). Frame findings using our new dynamical narrative.
- **Week 7:** *Drafting.* Consolidate findings, generate phase-plane visualizations, and finalize paper.

LLM Usage

LLMs (Gemini) were utilized during the drafting of this proposal for paragraph revision, academic tone polishing, and structuring the LaTeX document formatting.

References

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